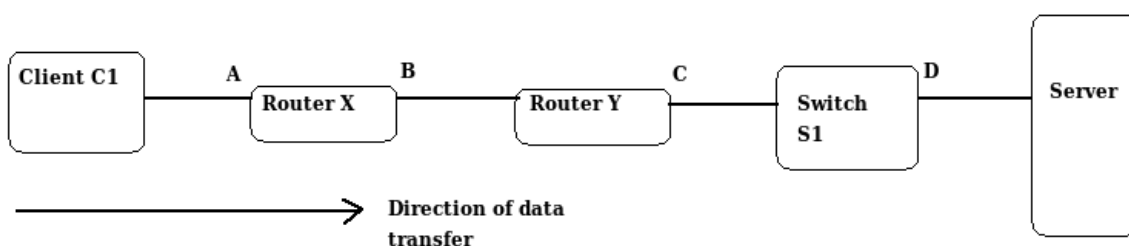


1. Consider the scenario shown below, with a client computer connected to a web-server via two routers and one switch. Client C1 wants to send data to Server. You are given four observation points (indicated A-D) in the corresponding Figure.

At each of these ingress/egress interfaces (A-D), work out the source and destination network layer (IP) and data link layer (MAC) address required to complete this data transfer.

Notes:

- Direction of data transfer to consider for this scenario: client C1 → server
- Represent network layer addresses by: IP – C1, IP-Server, IP-Router#, etc.
- Denote data link layer addresses using MAC-C1, MAC-Server, etc.



At Point A:

Source IP address: [e.g, IP-Router X]
Destination IP address:
Source MAC address: [e.g, MAC-Client C1]
Destination MAC address:

At Point B:

Source IP address:
Destination IP address:
Source MAC address:
Destination MAC address:

At Point C:

Source IP address:
Destination IP address:
Source MAC address:
Destination MAC address:

At Point D:

Source IP address:
Destination IP address:
Source MAC address:

Destination MAC address:

Roll no: _____

Q2. Explain: TCP helps the overall network to have less congested traffic. Which features of the TCP provide this service? Briefly explain the working of these features.

National University of Computer and Emerging Sciences, Lahore Campus



Course:
Program:
Duration:
Date:
Section:

Computer Networks
BS(Computer Science)
20 Minutes
Oct, 2019
E (b)

Course Code: CS307
Semester: Fall 2019
Total Marks: 14
Quiz: 2
Page(s): 1

Name _____ Roll No. _____

Question 1: [Marks 4]

1. Domain Name System (DNS), has two types of messages

- a) Query and Resolver
- b) Hosted and Generic
- c) Query and Responsive
- d) Resolver and Informative

2. Which multiplexing technique transmits digital signals?

- a) FDM
- b) TDM
- c) WDM
- d) FDM & WDM

3. Unreliable and connectionless protocol is

- a) TCP
- b) ARP
- c) RARP
- d) IP

4. Which of the following is false with respect to UDP

- a) Connection-oriented
- b) Unreliable
- c) Transport layer protocol
- d) All of the mentioned

Question 2: [Marks 4]

Indicate whether you think TCP or UDP would be better suited for each of the following applications and briefly explain why.

• Internet telephony voice channel

UDP – reliability is not needed, but a guaranteed transfer rate is. UDP will give a higher-quality transmission if there are limited network resources.

• IRC (chat) client/server

Definitely TCP – reliability, in-order delivery is needed, and bandwidth/latency guarantees are not.

Question 3: [Marks 6]

Consider the TCP connection mechanism.

- i. What mechanism is used to set up a TCP connection?
- ii. Using diagram show how the mechanism works.

National University of Computer and Emerging Sciences, Lahore Campus



Course:
Program:
Duration:
Date:
Section:

Computer Networks
BS(Computer Science)
20 Minutes
Oct, 2019
E (a)

Course Code: CS307
Semester: Fall 2019
Total Marks: 14
Quiz: 2
Page(s): 1

Name _____ Roll No. _____

Question 1: [Marks 4]

- 1. Which of the following is false with respect to UDP**
 - a) Connection-oriented
 - b) Unreliable
 - c) Transport layer protocol
 - d) All of the mentioned
- 2. Which multiplexing technique transmits digital signals?**
 - a) FDM
 - b) TDM
 - c) WDM
 - d) FDM & WDM
- 3. Domain Name System (DNS), has two types of messages**
 - a) Query and Resolver
 - b) Hosted and Generic
 - c) Query and Responsive
 - d) Resolver and Informative
- 4. Unreliable and connectionless protocol is**
 - a) TCP
 - b) ARP
 - c) RARP
 - d) IP

Question 2: [Marks 4]

Indicate whether you think TCP or UDP would be better suited for each of the following applications and briefly explain why.

State any assumptions that you are making for each application.

• Streaming video client/server

UDP – reliability is not needed, but a low latency is. UDP will give a higher-quality transmission if there are limited network resources.

• Multiplayer online first-person shooting game

Probably UDP – reliability is needed here, but so is low latency. A case could be made for both, but many games use UDP in order to achieve the low latency.

Question 3: [Marks 6]

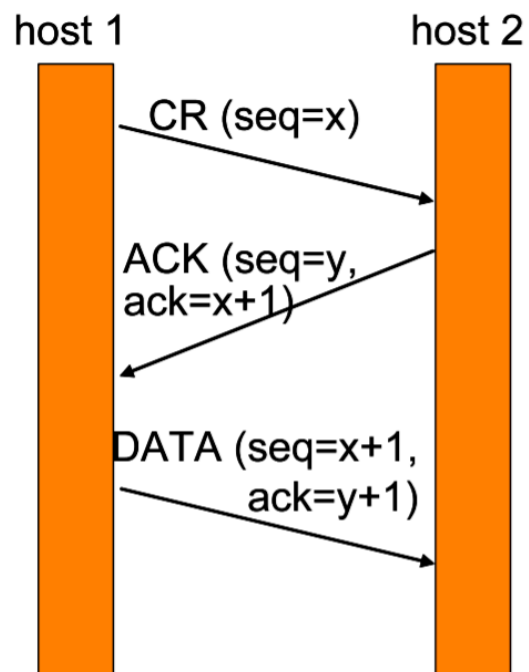
Consider the TCP connection mechanism.

- i. What mechanism is used to set up a TCP connection?

- ii. Using diagram show how the mechanism works.

Answer:

1. 3 way handshake
2. Explained in diagram:



National University of Computer and Emerging Sciences, Lahore Campus
Quiz2 [BSCS: Section C] Fall 2022

Computer Networks (Code: CS3001)

Quiz Date: September 21, 2022

Total Marks: 10

Duration: 12 -Minutes

Name ----- Roll #----- Section -----

Instructions: Provide your answer(s) on this sheet only. You can make use of rough sheet (do not attach to this sheet).

Q1: Consider the following string of ASCII characters that were captured by Wireshark when the browser sent an HTTP GET message (i.e., this the actual content of an HTTP GET message). The character are carriage return and line-feed characters (that is, the italicized character string in the text below represents the single carriage-return character that was contained at the point in the HTTP header). Answer the following questions (2*5 = 10)[CLO 2]

```
GET /cs453/index.html HTTP/1.1<cr><lf>Host: gai
a.cs.umass.edu<cr><lf>User-Agent: Mozilla/5.0 (
Windows;U; Windows NT 5.1; en-US; rv:1.7.2) Gec
ko/20040804 Netscape/7.2 (ax) <cr><lf>Accept:ex
t/xml, application/xml, application/xhtml+xml, text
/html;q=0.9, text/plain;q=0.8,image/png,*/*;q=0.5
<cr><lf>Accept-Language: en-us,en;q=0.5<cr><lf>Accept-
Encoding: zip,deflate<cr><lf>Accept-Charset: ISO
-8859-1,utf-8;q=0.7,*;q=0.7<cr><lf>Keep-Alive: 300<cr>
<lf>Connection:keep-alive<cr><lf><cr><lf>
```

- A. What is the URL of the document being requested by the browser?
<http://gaia.cs.umass.edu/cs453/index.html>
- B. What version of HTTP is the browser running?
The browser is running HTTP version 1.1, as indicated just before the first pair.
- C. Does the browser request a non-persistent or persistent connection?
The browser is requesting a persistent connection, as indicated by the Connection: keep-alive.
- D. What is the IP address of the host on which browser is running?
This is a trick question. This information is not contained in an HTTP message anywhere. So there is no way to tell this from looking at the exchange of HTTP messages alone. One would need information from the IP datagrams (that carried the TCP segment that carried the HTTP GET request) to answer this question.
- E. What type of browser initiates this message?
Mozilla/5.0

National University of Computer and Emerging Sciences, Lahore Campus
Quiz2 [BSCS: Section D] Fall 2022

Computer Networks (Code: CS3001)

Quiz Date: September 21, 2022

Total Marks: 10

Duration: 15 -Minutes

Name ----- Roll #----- Section -----

Instructions: Attempt all questions on this sheet. You can make use of rough sheet (do not attach to this sheet).

Q1: We observe an increase in the total delay (time) for those requests that encounter a web cache (a penalty because of the involvement of the cache). Let LD be the average LAN delay, ALD be the access link delay, ID be the Internet delay, and RT be the total response time. Moreover, assume that ACP is the average cache penalty delay. The following values are given LD = 10ms, ALD = 50ms, ID = 100ms, and ACP = 20 ms. Considering the above scenario and given values, answer the following questions: (4 + 6 = 10 Marks) **[CLO 4]**

(A) Find the average response time, TR, without use of cache?

Solution:

No cache: average response time = TR = LD + ALD + ID, thus RT = 10 + 50 + 100 = 160ms

(B) Find the average response delay (time) when a cache is used and the hit ratio is 80%?

Solution: Solution: Cache and hit time + Cache and miss time i.e.:

$RT = [0.8 * (LD + ACP)] + [0.2 * (LD + ALD + ID + ACP)]$

$RT = [0.8 * (10 + 20)] + [0.2 * (10 + 50 + 100 + 20)] = 0.8 \times 30 + 0.2 \times 180 = 24 \text{ ms} + 36 \text{ ms} = 60 \text{ ms}$

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2023
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	21-September-2023	Section	5J
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 2]

1. A socket consists of:
 - a) MAC Address
 - b) IP Address
 - c) Port Number
 - d) Both (b) & (c)
2. Multiple objects can be sent over a same TCP connection between client and server in:
 - a) Non-Persistent HTTP
 - b) Persistent HTTP
 - c) None
 - d) Both (a) & (b)
3. HTTP is a _____ protocol?
 - a) Push
 - b) Pull
 - c) Stateless
 - d) Both (b) & (c)

True/False:

1. HTTP client requests by establishing a UDP connection to a particular port on the server [T / F]
2. SMTP uses port 29. [T / F]

Q3.

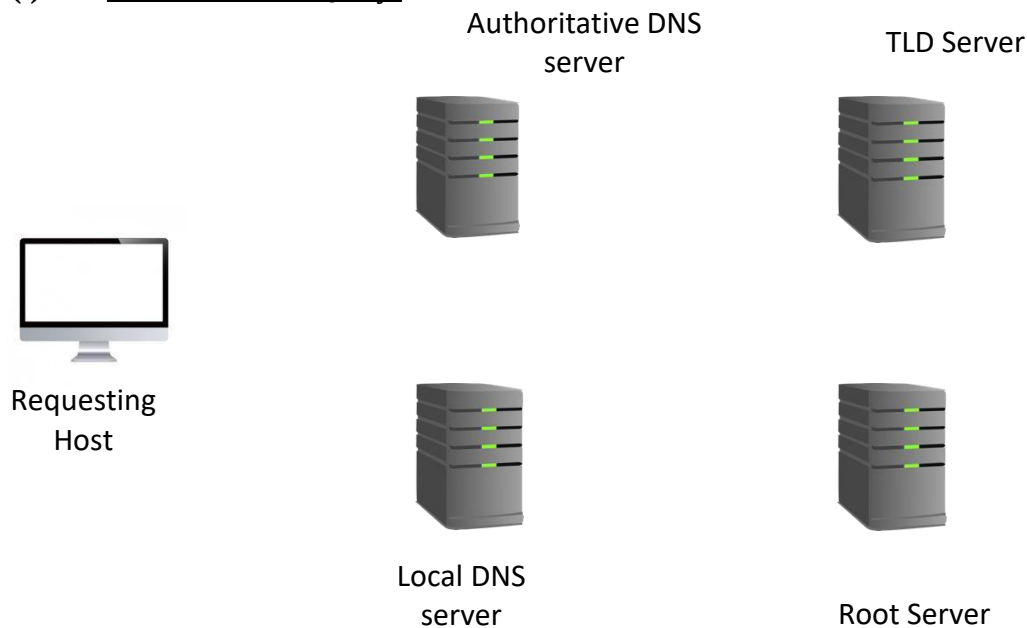
(a) What is RTT?

[2 Marks] [CLO 2]

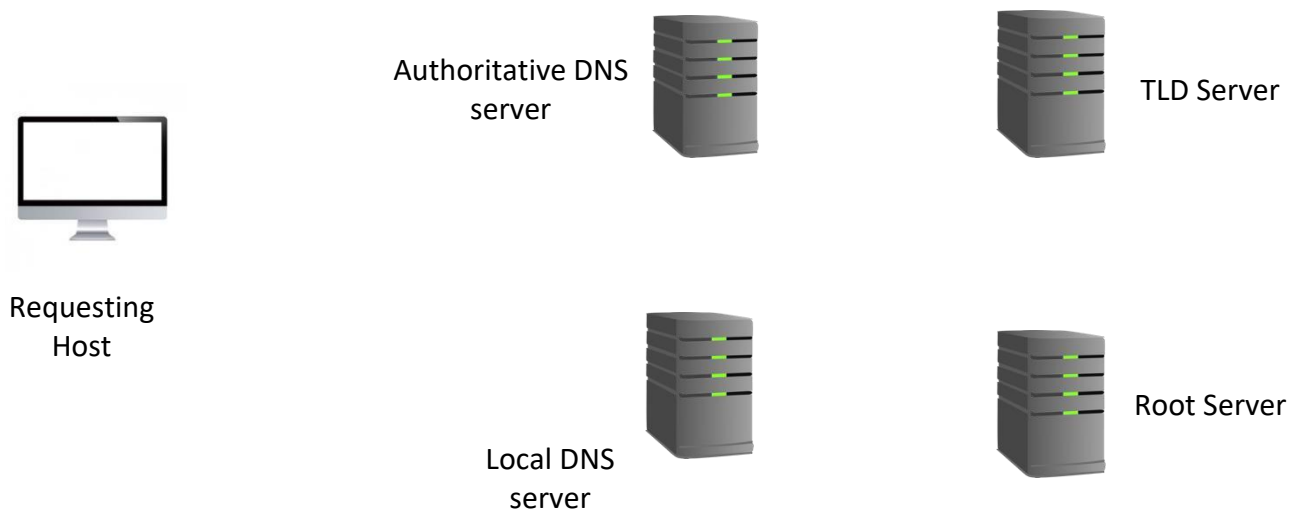
(b) Elaborate the difference between Recursive and Iterative DNS query with help of the figures below? [4 + 4 Marks] [CLO 2]

NOTE: Draw arrow lines showing messages between the requesting host/ DNS servers (& vice versa) to complete a DNS query/ response sequence. Furthermore, number each line to show the communication sequence.

(i) Recursive DNS Query:



(ii) Iterative DNS Query:



Solution Quiz#2(5J)

MCQ's

1. d
2. b
3. d

T/F

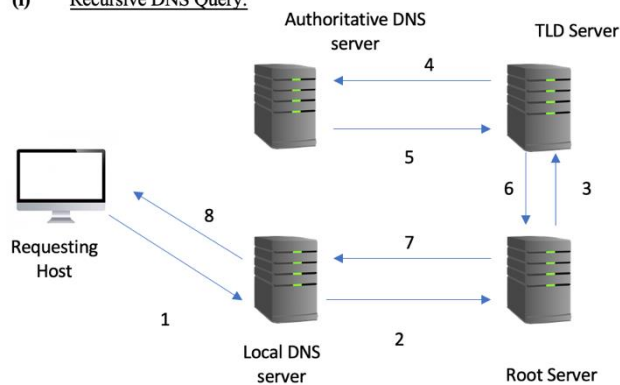
1. F
2. F

Q2.(a)

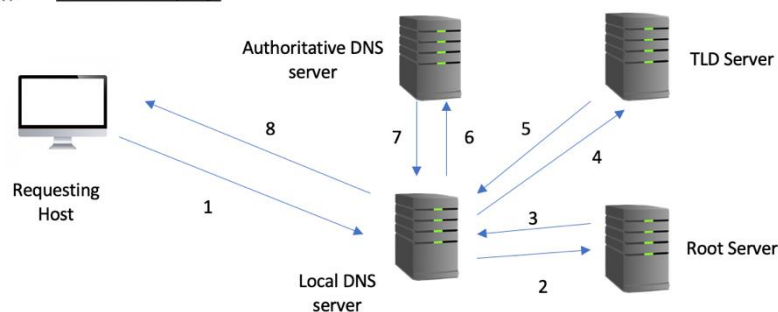
RTT is the round trip time taken for an HTTP message to reach the server and back.

Q2.(b)(i) & (ii)

(i) Recursive DNS Query:



(i) Iterative DNS Query:



	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2023
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	21-September-2023	Section	5K
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 2]

1. When displaying webpage, the application layer uses the _____ protocol.
 - a) FTP
 - b) HTTP
 - c) TCP
 - d) POP3
2. Multiple objects can be sent over a same TCP connection between client and server in:
 - a) Non-Persistent HTTP
 - b) Persistent HTTP
 - c) None
 - d) Both (a) & (b)
3. The underlying transport layer protocol used by SMTP is:
 - a) POP3
 - b) UDP
 - c) TCP
 - d) IMAP

True/False:

1. Socket is only a combination of port number. [T / F]
2. IMAP maintains state across sessions. [T / F]

Q2. Suppose you are running a webserver on your home machine that connects to the network service provider through a DSL link with 100 Kbps ($1\text{K bits} = 10^3 \text{ bits}$) connection speed. You come to the University and try to access a web page on the webserver running on your home machine. The campus LAN in the University runs at 100 Mbps with an access link of 8 Mbps ($1\text{M bits} = 10^6 \text{ bits}$). Further assume that the RTT from your University campus to your home is 150 ms. The base html file size you request is 10 KB (10 kilo bytes). This base file also includes 3 referenced files (3 small images.) The first image that appears on the page is 20 KB (20 kilo bytes, the second which appears somewhere in the middle of the pages is 30 KB (30 kilo bytes) and the third image, towards the end of the page, is 15 KB (15 kilo bytes.) Requests for these images are sent by your browser in that order, without using any Parallel TCP connections. (Neglect the time spent in processing of html files, i.e., processing and queuing delays.) **[10 Marks] [CLO 2]**

- (a) What is the total number of HTTP requests generated from the browser on campus computer?

- (b) How long would it take your browser from sending the first HTTP request to receiving the base html file?

- (c) Assume http/1.0 (non-persistent). How long would it take your browser from sending the request to displaying the complete web page?

Solution Quiz#2(5K)

MCQ's

1. b
2. b
3. c

T/F

1. F
2. T

Q2. Solution:-

(a) Non-persistent = 4 HTTP requests (One for each of the four objects), persistent without pipelining = 4 HTTP requests, persistent with pipelining = 2 HTTP requests

(b) $2RTT + T = 2RTT + [L / \min(R_{DSL}, R_{LAN}, R_{ACCESS\ LINK})]$ i.e. $(2 * 150ms) + (10K * 8 / 100K) = 300ms + 800ms = 1100ms = 1.1\ s$

(c) $(4 * 2 RTT) + T = 8 RTT + (L_{BASE} / \min(R_{DSL}, R_{LAN}, R_{ACCESS\ LINK})) + [L_{1ST\ object} / \min(R_{DSL}, R_{LAN}, R_{ACCESS\ LINK})] + [L_{1ST\ object} / \min(R_{DSL}, R_{LAN}, R_{ACCESS\ LINK})] + [L_{1ST\ object} / \min(R_{DSL}, R_{LAN}, R_{ACCESS\ LINK})]$ i.e. $[4 * 2 * 150ms] + [(10K + 20K + 30K + 15K) * 8 / 100k] = 7200ms = 7.2\ s$

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National University of Computer and Emerging Sciences, Lahore
Campus Quiz2 [BCS: Section 5A] Fall 2024

Computer Networks (Code: CS3001)

Quiz Date: September 19, 2024

Total Marks: 13

Duration: 20 -Minutes

Name _____

Roll # _____

Instructions: Attempt all questions on this sheet. You can make use of rough sheet (do not attach to this sheet).

Q1: Identify and encircle the correct option(s).

(3 Marks)

[CLO 1]

1. With P2P file sharing, the peer that is distributing the file is labeled as:

(a) Client (b) Server (c) router (d) None of these

2. HTTP uses _____ connections in its default mode:

(a) persistent (b) parallel (c) non-persistent (d) None of these

3. DNS runs over TCP and uses port 53.

(a) TCP, 23 (b) UDP, 23 (c) TCP, 53 (d) UDP, 23

→ Though peers act as both client and server, in a single comm. session it only acts as either a server or a client

Q2: We observe an increase in the total delay (time) for those requests that encounter a web cache (a penalty because of the involvement of the cache). Let LD be the average LAN delay, ALD be the access link delay, ID be the Internet delay, and RT be the total response time. Moreover, assume that ACP is the average cache penalty delay. The following values are given LD = 10ms, ALD = 50ms, ID = 100ms, and ACP = 20 ms. Considering the above scenario and given values, answer the following questions: (4 + 6 = 10 Marks) [CLO 2]

(A) Find the average response time, TR, without use of cache?

(B) Find the average response delay (time) when a cache is used, and the cache hit ratio is 80%?

a) $RT = LD + ALD + ID = 10 + 50 + 100 = 160 \text{ ms}$

hit rate
b) $[HR \times (LD + ACP)] + [Miss \text{ rate} \times RT]$
 $[0.8 \times (10 + 20)] + [0.2 \times 160]$
 $= 56 \text{ ms}$

Assumption: Web cache deployed in a LAN

SOLUTION QUIZ-02 BCS-5A

QUESTION-01

1. C
2. A
3. B

TRUE/FALSE

- i. False
- ii. False

QUESTION-02

- Fill in the names of the layers, a protocol that corresponds to each layer and the name of the data unit in each one. Then, place the appropriate letter in the blank associated with the layer for the proper description of its services below as illustrated in the solved examples.

	Name of the layer	Protocols	Name of data unit	Proper description
Layer 5	application	HTTP,smtp, dns,	message	d
Layer 4	transport	UDP, TCP	segment	c
Layer 3	Network or internet	IP	Packet or datagram	b
Layer 2	Data link	Ethernet	frame	e
Layer 1	Physical layer	Analog/digital	bit	a

- a) Provides a definition of mechanical and electrical standards for communication system
- b) Concerned with routing packets across a communication network
- c) Provides end-to-end logical communication between application processes.
- d) Format and transfer files between communication paradigm and the user's software
- e) Frames of data are transferred across a single link

SOLUTION QUIZ-02 BCS-5C

QUESTION-01

1. C
2. D
3. A

TRUE/FALSE

- i. False
- ii. False

QUESTION-02

1. $RTT = \text{propagation delay} * 2 = 10 \text{ ms} * 2 = 20 \text{ ms} = 0.02 \text{ sec}$
 $RTT \text{ count} = 2 \text{ RTT (for base object)} + 10 * 2 \text{ RTT (for 10 reference images)} = 22 \text{ RTTs} = 22 * 0.02 = 0.44 \text{ sec}$
 $\text{Base file transmission time} = L/R = 50 \text{ Kbytes} / 2 \text{ Mbps} = (50 * 10^3 * 8) / (2 * 10^6) = 0.2 \text{ sec}$
 $10 \text{ reference images transmission time} = 10 (L/R) = 10 * [25 \text{ Kbytes} / 2 \text{ Mbps} = (25 * 10^3 * 8) / (2 * 10^6) = 1 \text{ sec}]$
 $\text{Total time} = 22 \text{ RTT time} + \text{base file transmission} + \text{ten reference images transmission times} = 0.44 + 0.2 + 1.0 = \mathbf{1.64} \text{ seconds.}$
2. $RTT = \text{propagation delay} * 2 = 10 \text{ ms} * 2 = 20 \text{ ms} = 0.02 \text{ sec}$
 $RTT \text{ count} = 1 \text{ RTT to open TCP connection} + 11 \text{ RTT (to request 1 base + 10 reference objects)} = 12 \text{ RTT} = 12 * 0.02 = 0.24 \text{ sec}$
 $\text{File transmission times are the same for base and reference images as in part 1.}$
 $\text{Base file transmission time} = L/R = 50 \text{ Kbytes} / 2 \text{ Mbps} = (50 * 10^3 * 8) / (2 * 10^6) = 0.2 \text{ sec}$
 $10 \text{ reference images transmission time} = 10 (L/R) = 10 * [25 \text{ Kbytes} / 2 \text{ Mbps} = (25 * 10^3 * 8) / (2 * 10^6) = 1 \text{ sec}]$
 $\text{Total time} = 12 \text{ RTT time} + \text{base file transmission} + \text{ten reference images transmission times} = 0.24 + 0.2 + 1.0 = \mathbf{1.44} \text{ seconds.}$
3. $RTT = \text{propagation delay} * 2 = 10 \text{ ms} * 2 = 20 \text{ ms} = 0.02 \text{ sec}$
 $RTT \text{ count} = 1 \text{ RTT to open TCP connection} + 1 \text{ RTT to request base file} + 1 \text{ RTT to request all reference objects} = 3 \text{ RTT} = 3 * 0.02 = 0.06 \text{ sec}$
 $\text{File transmission times are the same for base and reference images as in part 1 or 2.}$
 $\text{Base file transmission time} = L/R = 50 \text{ Kbytes} / 2 \text{ Mbps} = (50 * 10^3 * 8) / (2 * 10^6) = 0.2 \text{ sec}$
 $10 \text{ reference images transmission time} = 10 (L/R) = 10 * [25 \text{ Kbytes} / 2 \text{ Mbps} = (25 * 10^3 * 8) / (2 * 10^6) = 1 \text{ sec}]$
 $\text{Total time} = 3 \text{ RTT time} + \text{base file transmission} + \text{ten reference images transmission times} = 0.06 + 0.2 + 1.0 = \mathbf{1.26} \text{ seconds.}$

**National University of Computer and Emerging Sciences, Lahore
Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	18-September-2025	Section	5 A , C
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 2]

1. When retrieving emails from a recipient's mail server, which protocol is commonly used?
 - a) SMTP
 - b) DNS
 - c) IMAP
 - d) FTP
2. Which of the following is present in both, an HTTP request line & HTTP response status line?
 - a) HTTP Version Number
 - b) URL
 - c) Method
 - d) None of the above
3. Which DNS record type is used to map an alias (alternate name) to a canonical domain name?
 - a) A
 - b) CNAME
 - c) MX
 - d) PTR

True/False:

1. Socket number is the combination of only port numbers. [T / F]
2. Cookie information is stored only on the server-side in a computer network.. [T / F]

Q2:

[10 Marks] [CLO 2]

Fill in the blanks for each layer of the 5-layer Internet model. For every layer, specify each value **(0.5 marks for each)**:

	Name of the Layer	Protocol (Any One Example)	Name of the Data Packet	Description from below (A to E)
Layer 7				
Layer 4				
Layer 3				
Layer 2				
Layer 1				

Descriptions:

- A.** Provides a definition of mechanical and electrical standards for communication systems
- B.** Concerned with routing packets across a communication network
- C.** Provides end-to-end logical communication between application processes
- D.** Format and transfer files between communication paradigm and the user's software
- E.** Frames of data are transferred across a single link

**National University of Computer and Emerging Sciences,
Lahore Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	18-September-2025	Section	5 A , C
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 2]

1. Which HTTP header is commonly used in Conditional GET requests to check if a resource has been modified since a certain date?
 - a) Cache-Control
 - b) Last-Modified
 - c) If-Modified-Since
 - d) Expires
2. What is the primary purpose of an MX record in DNS?
 - a) used to map an alias to a canonical domain name
 - b) specifies which DNS server is authoritative for this domain
 - c) used to map (point) a domain name to an IP - address
 - d) mail server aliasing
3. In the context of email communication, which protocol is commonly used for sending emails between servers?
 - a) SMTP
 - b) DNS
 - c) IMAP
 - d) HTTP

True/False:

1. DNS uses TCP as a transport layer protocol. [T / F]
2. Persistent Pipelining HTTP connections require a new connection to be established for each resource request. [T / F]

Q2:

[3 + 3 + 4 Marks] [CLO 2]

Alice is shopping online via the internet with a connection of 2 Mbps. The base web page of size 50 Kbytes contains 10 images of 25 Kbytes each. Assume that the one way propagation delay is 10 ms.

1. How long does it take for the entire page (including images) to be completely displayed on Alice's screen, assuming **non-persistent HTTP using a single connection** at a time? Show your calculation in detail.
2. What would be the total time for the complete page including base and images to appear on Alice's screen if she used **persistent HTTP without pipelining** instead of non-persistent HTTP? Show your calculation in detail.
3. Repeat part (2) but use **Persistent HTTP with Pipelining**.

SOLUTION QUIZ-02 BCS-6A

QUESTION-01

- 1) C
- 2) D
- 3) A

TRUE/FALSE

- i. False
- ii. False

QUESTION-02

a) $RTT = 2 \times \text{one-way propagation delay} = 5 \text{ ms} = 0.005$

Time to download the web page with non-persistent HTTP:

42 RTT + transmission of the base file (20 Kbytes) + time transmission of the 20 images (15 Kbytes each)

$$= 42 \times 0.005 + (8 \times 20 \times 10^3 / 0.5 \times 10^6) + (20 \times 8 \times 15 \times 10^3 / 0.5 \times 10^6)$$

$$= 0.21 + 0.32 + 4.8 = 5.33 \text{ sec}$$

b) Time to download the web page with persistent HTTP without pipelining

22RTT + transmission time of the base file (20 Kbytes) + time transmission of the 20 images (15kBytes)

$$= (22 \times 0.005) + (8 \times 20 \times 10^3 / 0.5 \times 10^6) + (20 \times 8 \times 15 \times 10^3 / 0.5 \times 10^6)$$

$$= 0.11 + 0.32 + 4.8 = 5.23 \text{ sec}$$

c) Time to download the web page with persistent HTTP with pipelining

= 3 RTT + time transmission of the base file (20 Kbytes) + time transmission of the one image (15 Kbytes)

$$= 3 \times 0.005 + (8 \times 20 \times 10^3 / 0.5 \times 10^6) + (8 \times 15 \times 20 \times 10^3 / 0.5 \times 10^6)$$

$$= 0.015 + 0.32 + 4.8$$

$$= 5.135 \text{ sec}$$

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2024
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	22-Feb-2024	Section	6A
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 2]

1. Which HTTP header is commonly used in Conditional GET requests to check if a resource has been modified since a certain date?
 - a) Cache-Control
 - b) Last-Modified
 - c) If-Modified-Since
 - d) Expires
2. What is the primary purpose of an MX record in DNS?
 - a) used to map an alias to a canonical domain name
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 - c) used to map (point) a domain name to an IP - address
 - d) mail server aliasing
3. In the context of email communication, which protocol is commonly used for sending emails between servers?
 - a. SMTP
 - b. DNS
 - c. IMAP
 - d. HTTP

True/False:

1. DNS uses TCP as a transport layer protocol [T / F]
2. Persistent with Pipelining HTTP connections require a new connection to be established for each resource request [T / F]

Question 2: [Marks 10]

Alice is shopping online via internet with a connection of **0.5 Mbps**. The web page of size **20 Kbytes** contains **20 images of 15 Kbytes** each. Assume that the **one way propagation delay** is **2.5 ms**. **[3 + 3 + 4 Marks] [CLO 2]**

- a) How long does it take for the page (including images) to appear on Alice's screen, assuming **non-persistent HTTP** using a single connection at a time? Show your calculation in details.
- b) What would become the time for the page including images) to appear on Alice's screen if she used **persistent HTTP without pipelining** instead of non-persistent HTTP? Show your calculation in details.
- c) Repeat part b) **if persistent HTTP with pipelining** is used. Again, show your calculation in details.

SOLUTION QUIZ-02 BSE-6A

QUESTION-01

- 1) C
- 2) A
- 3) B

TRUE/FALSE

- i. False
- ii. False

QUESTION-02

- a) $RTT1 = 2 * d1/s1$ $RTT2 = 2 * d2/s2$
- b) Delay = $RTT1$ (connection with server1) + $RTT1$ (request and response for cong415.html) + L/R +
 $RTT1$ (connection with server1) + $RTT1$ (request and response for image1.jpg) + $L1/R$ +
 $RTT1$ (connection with server1) + $RTT1$ (request and response for image2.jpg) + $L2/R$ +
 $RTT2$ (connection with server2) + $RTT2$ (request and response for image3.jpg) + $L3/R$ +
 $RTT2$ (connection with server2) + $RTT2$ (request and response for image4.jpg) + $L4/R$.
- c) Delay = $RTT1$ (connection with server1) + $RTT1$ (request and response for cong415.html) + L/R +
 $RTT1$ (request and response for image1.jpg) + $L1/R$ +
 $RTT1$ (request and response for image2.jpg) + $L2/R$ +
 $RTT2$ (connection with server2) + $RTT2$ (request and response for image3.jpg) + $L3/R$ +
 $RTT2$ (request and response for image4.jpg) + $L4/R$.

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2024
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	22-Feb-2024	Section	BSE-6A
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 2]

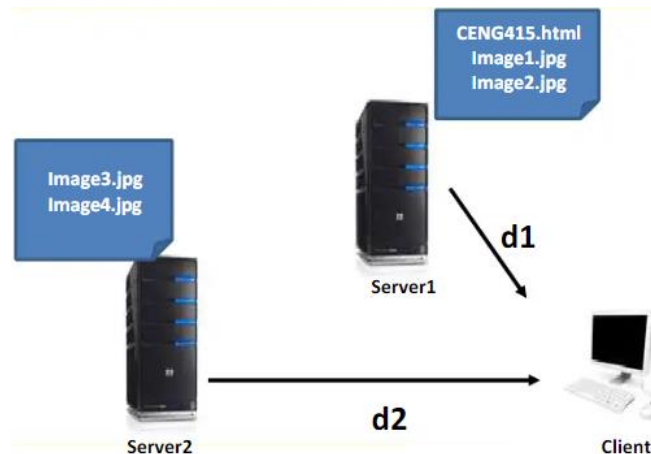
1. When retrieving emails from a recipient's mail server, which protocol is commonly used?
 - a. SMTP (Simple Mail Transfer Protocol)
 - b. DNS
 - c. IMAP (Internet Message Access Protocol)
 - d. HTTP (Hypertext Transfer Protocol)
2. Which of the following is present in both an HTTP request line and a status line?
 - a) HTTP version number
 - b) URL
 - c) Method
 - d) None of the mentioned
3. Which DNS record type is used to map an alias (alternate name) to a canonical domain name?
 - a. A
 - b. CNAME
 - c. MX
 - d. PTR

True/False:

- i. Socket number is the combination of only port number. [T / F]
- ii. Cookie information is stored only on the server-side in a computer network. [T / F]

Q2. An HTTP file ceng415.html with size L is located at server1, this file references 4 images: Image 1.jpg (size L_1), Image2.jpg (size L_2), Image3.jpg (size L_3) and Image4.jpg (size L_4). The first 2 images are stored in server1, but Image3 and Image4 are stored in another server: server2. Server1 is located at a distance d_1 while server2 is at distance d_2 far away from the client. [2+4+4=10 marks] [CLO 2]

Assume that we have a constant bit rate R and a line speed S over all the links.
In terms of L , L_1 , L_2 , L_3 , L_4 , d_1 , d_2 , R and S , answer the following questions.



- Find the Round Trip Times **RTT1** between the client and server1 and **RTT2** between the client and server2.
- If we are using a **non-persistent http protocol**, compute the total time that it would take the browser on the client to retrieve the entire document.
- Re-calculate the total time to retrieve the entire document, if the client browser is now using **persistent http protocol without pipelining**.

SOLUTION QUIZ-02 BSE-6A

QUESTION-01

- 1) B
- 2) B
- 3) C
- 4) B

TRUE/FALSE

- i. True
- ii. False

QUESTION-02

(a) Non-persistent = 4 HTTP requests (One for each of the four objects), persistent without pipelining = 4 HTTP requests, persistent with pipelining = 2 HTTP requests

(b) $2RTT + T = 2RTT + [L / \min(R_{DSL}, R_{LAN}, R_{ACCESS\ LINK})]$ i.e. $(2 * 150ms) + (10K \times 8 / 100K) = 300ms + 800ms = 1100ms = 1.1\ s$

(c) $(4 * 2 RTT) + T = 8 RTT + (L_{BASE} / \min(R_{DSL}, R_{LAN}, R_{ACCESS\ LINK})) + [L_{1ST\ object} / \min(R_{DSL}, R_{LAN}, R_{ACCESS\ LINK})] + [L_{1ST\ object} / \min(R_{DSL}, R_{LAN}, R_{ACCESS\ LINK})] + [L_{1ST\ object} / \min(R_{DSL}, R_{LAN}, R_{ACCESS\ LINK})]$ i.e. $[4 * 2 \times 150ms] + [(10K + 20K + 30K + 15K) \times 8 / 100k] = 7200ms = 7.2\ s$

SOLUTION QUIZ-02 BSE-6C

QUESTION-01

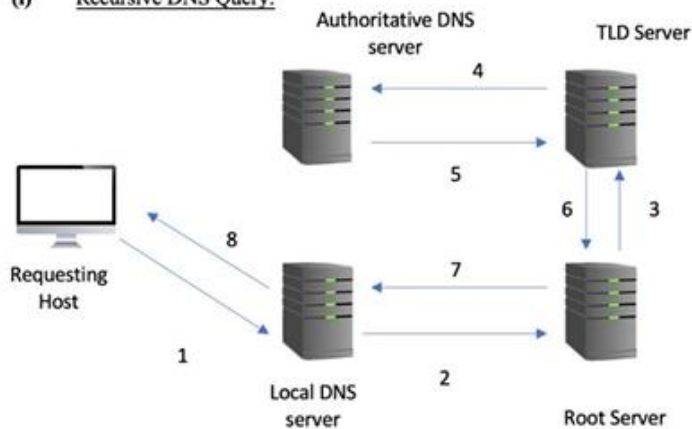
- a) B. SMTP
- b) B. Recursive queries minimize the burden on TLD servers, while iterative queries require multiple requests by the client.
- c) C. HTTP/1.1
- d) C. The resolver must query higher-level DNS servers to refresh the record.

TRUE/FALSE

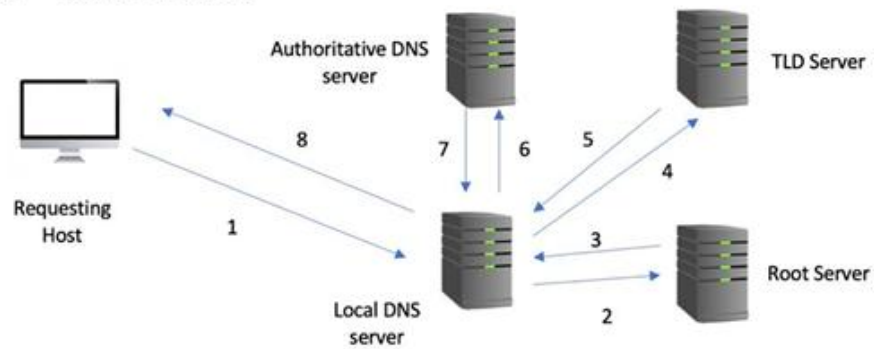
- i. True
- ii. True

QUESTION-02

(i) Recursive DNS Query:



(i) Iterative DNS Query:



National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (SE)	Semester:	Spring 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	20-February-2025	Section	6A
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[6 marks] [CLO 2]

- The protocol commonly responsible for sending emails from a client to a server is:
 - IMAP
 - SMTP
 - POP3
 - HTTP
- What is the primary purpose of cookies in HTTP?
 - To reduce DNS queries
 - To store client-side session data
 - To enable encryption
 - To optimize file transfer speeds
- Which statement about persistent HTTP is correct?
 - Persistent HTTP creates a new TCP connection for each request.
 - Persistent HTTP requires only one RTT to download a webpage with multiple objects.
 - Persistent HTTP reuses the same TCP connection for multiple requests.
 - Persistent HTTP is slower than non-persistent HTTP.
- What does the 304 Not Modified HTTP status code indicate?
 - The requested resource could not be found.
 - The requested resource was not modified since the last request.
 - The server encountered an internal error.
 - The requested resource was successfully retrieved.

True/False:

- HTTP is a stateless protocol, meaning it does not retain any information about client requests once the server responds. [T / F]
- A web page containing text and three images will require one HTTP request from the client and one HTTP response from the server. [T / F]

Question 2:**[Marks 9] [CLO 2]**

Suppose you are running a webserver on your home machine that connects to the network service provider through a DSL link with 100 Kbps ($1\text{K bits} = 10^3 \text{ bits}$) connection speed. You come to the University and try to access a web page on the webserver running on your home machine. The campus LAN in the University runs at 100 Mbps with an access link of 8 Mbps ($1\text{M bits} = 10^6 \text{ bits}$). Further assume that the RTT from your University campus to your home is 150 ms. The base html file size you request is 10 KB (10 kilo bytes). This base file also includes 3 referenced files (3 small images.) The first image that appears on the page is 20 KB (20 kilo bytes, the second which appears somewhere in the middle of the page is 30 KB (30 kilo bytes) and the third image, towards the end of the page, is 15 KB (15 kilo bytes.) Requests for these images are sent by your browser in that order, without using any Parallel TCP connections. (Neglect the time spent in processing of html files, i.e., processing and queuing delays.)

- (a) What is the total number of HTTP requests generated from the browser on campus computer?
- (b) How long would it take your browser from sending the first HTTP request to receiving the base html file?
- (c) Assume http/1.0 (non-persistent). How long would it take your browser from sending the request to displaying the complete web page?

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (SE)	Semester:	Spring 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	20-February-2025	Section	6C
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[6 marks] [CLO 2]

- The protocol commonly responsible for sending emails from a client to a server is:
 - IMAP
 - SMTP
 - POP3
 - HTTP
- What is the main difference between recursive and iterative DNS queries?
 - Recursive queries minimize the burden on the client, while iterative queries burden the TLD server.
 - Recursive queries minimize the burden on TLD servers, while iterative queries require multiple requests by the client.
 - Recursive queries are faster than iterative queries.
 - There is no difference.
- Which HTTP version introduced pipelining to improve efficiency?
 - HTTP/0.9
 - HTTP/1.0
 - HTTP/1.1
 - HTTP/2
- What happens when a local DNS cache entry expires?
 - The query is redirected to the root server.
 - The DNS query fails permanently.
 - The resolver must query higher-level DNS servers to refresh the record.
 - The cache entry is automatically renewed.

True/False:

- DNS uses TCP for queries that exceed 512 bytes. **[T / F]**
- Non-persistent HTTP establishes a new TCP connection for every request-response pair. **[T / F]**

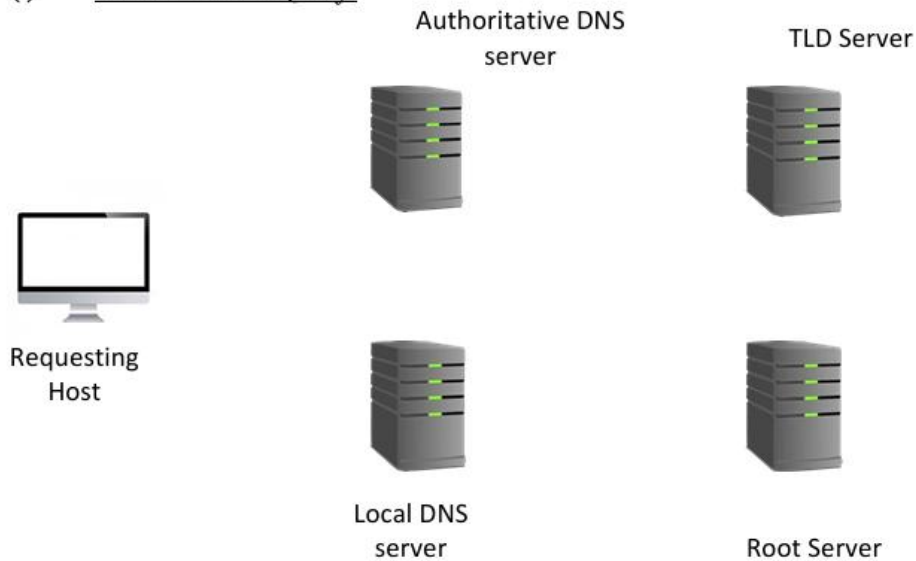
Question 2:

[Marks 9] [CLO 2]

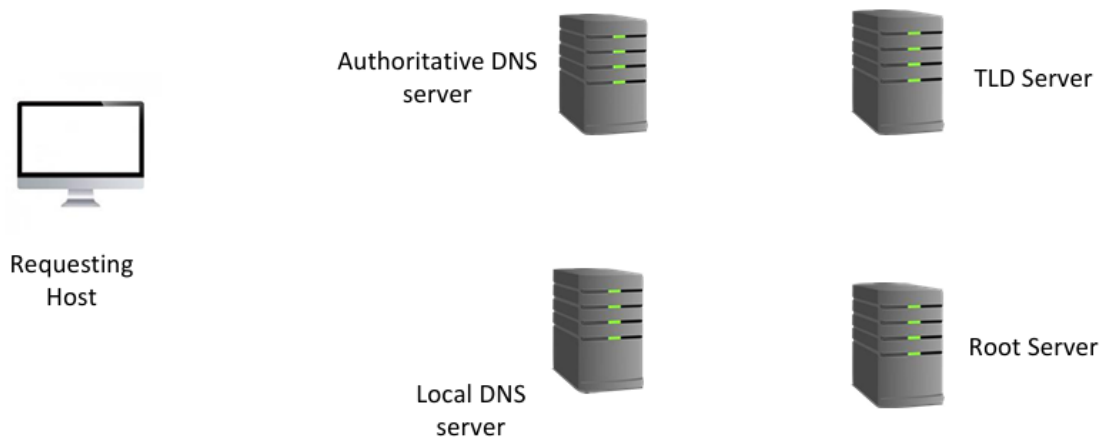
Elaborate the difference between Recursive and Iterative DNS query with help of the figures below,

NOTE: Draw arrow lines showing messages between the requesting host/ DNS servers (& vice versa) to complete a DNS query/ response sequence. Furthermore, number each line to show the communication sequence.

(i) Recursive DNS Query:



(ii) Iterative DNS Query:



National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:		Section	6B1
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

Q1. Carefully read the Question before attempting:

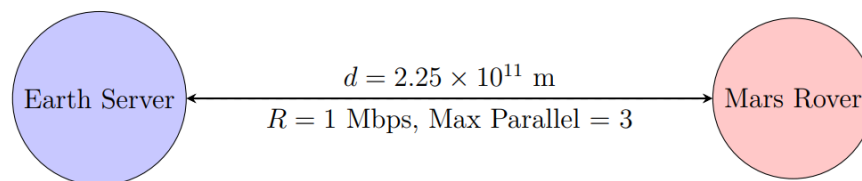
[15 marks]

An automated rover on Mars needs to download a firmware update from a server on Earth. The update consists of one base HTML index file (size $L_b = 2$ KB) and 8 referenced binary objects (each size $L_o = 100$ KB). The link characteristics are:

- Distance Earth-Mars: $d = 2.25 \times 10^{11}$ meters.
- Transmission Rate: $R = 1$ Mbps.

The rover opens parallel TCP connections but is limited to a maximum of $K = 3$ parallel connections due to memory constraints. Calculate the total time elapsed from the initial connection request until the last object is fully received. Assume:

- Short control packets (SYN, ACK, HTTP GET) have negligible transmission time compared to propagation delay.
- We are using Non-Persistent HTTP over the parallel connections. No DNS lookup is required (IP is hardcoded). Speed of light $c = 3 \times 10^8$ m/s.



National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:		Section	6B1
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

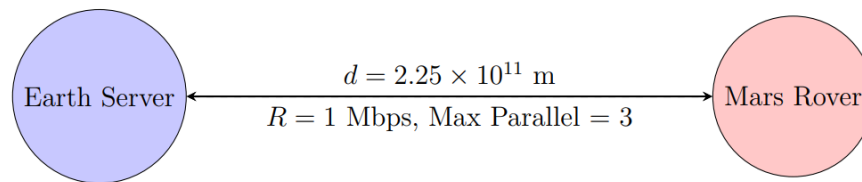
Q1. Carefully read the Question before attempting: [15 marks]

An automated rover on Mars needs to download a firmware update from a server on Earth. The update consists of one base HTML index file (size $L_b = 2$ KB) and 8 referenced binary objects (each size $L_o = 100$ KB). The link characteristics are:

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- We are using Non-Persistent HTTP over the parallel connections. No DNS lookup is required (IP is hardcoded). Speed of light $c = 3 \times 10^8$ m/s.



Solution:

1. Calculations:

$$d_{prop} = \frac{2.25 \times 10^{11}}{3 \times 10^8} = 750 \text{ s} \implies RTT = 2 \times d_{prop} = 1500 \text{ s}$$

$$d_{trans(base)} = \frac{2 \text{ KB} \times 8}{1 \text{ Mbps}} = 0.016 \text{ s}, \quad d_{trans(obj)} = \frac{100 \text{ KB} \times 8}{1 \text{ Mbps}} = 0.8 \text{ s}$$

2. Step 1: Base File Download (Non-Persistent: 2 RTT + Trans)

$$T_{base} = 2 \times RTT + d_{trans(base)} = 3000 + 0.016 = 3000.016 \text{ s}$$

3. Step 2: 8 Objects with K=3 Parallel Connections We download objects in batches.

Since $K = 3$, we need $\lceil 8/3 \rceil = 3$ batches. Note: Bandwidth is shared, so transmission time for 3 objects is $3 \times d_{trans(obj)}$.

- Batch 1 (Obj 1-3):** $2RTT + 3 \times 0.8 = 3000 + 2.4 = 3002.4 \text{ s}$
- Batch 2 (Obj 4-6):** $2RTT + 3 \times 0.8 = 3000 + 2.4 = 3002.4 \text{ s}$
- Batch 3 (Obj 7-8):** $2RTT + 2 \times 0.8 = 3000 + 1.6 = 3001.6 \text{ s}$

4. Total Time:

$$T_{total} = 3000.016 + 3002.4 + 3002.4 + 3001.6 = \mathbf{12006.416 \text{ s}}$$

**National University of Computer and Emerging Sciences,
Lahore Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	19-February-2026	Section	6A , 6B
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

Q1. Consider an **HTTP/2** connection downloading 3 objects: Object A (three frames, i.e. A1, A2, A3), Object B (one frame i.e. B1) and Object C (one frame i.e. C1). Each frame takes 10 ms to transmit.
[2 + 3 = 5 Marks] [CLO 1]

a) List the sequence of frame arrival at the client if Round Robin (FCFS with preemption) scheduling is used to prevent Head-of-Line (HOL) blocking.

b) Contrast this with HTTP/1.1 where Object A must be fully downloaded before B and C. How much earlier does Object B finish in HTTP/2 compared to HTTP/1.1? (i.e. calculate the time taken for Object B to be fully downloaded in HTTP/1.1 compared to in HTTP/2.)

Q2. We notice an increase in the total delay (time) for those requests that encounter a web cache (a penalty because of the involvement of the cache). Let L be the average LAN delay, A be the access link delay, I be the Internet delay, and R be the total response time. Also assume that P is the average cache penalty delay. Given that $L = 10$ ms, $I = 180$ ms, $P = 40$ ms, and $A = 60$ ms. **[5 + 5 = 10 Marks] [CLO 1]**

(i) Find the average response delay (time), R , when no cache is used?

(ii) Find the average response delay (time) when a cache is used and the hit ratio is 80%?

SOLUTION QUIZ-02 BSE-6A

QUESTION-01

- a. **Sequence:** Frame A1, Frame B1, Frame C1, Frame A2, Frame A3.
- b. **Comparison:**
 - a. **HTTP/1.1:** Order is A (all frames) -> B -> C.
 - i. B finishes after A completes (30ms) + B (10ms) = 40ms.
 - b. **HTTP/2:**
 - i. B finishes at slot 2 (20ms).
 - c. **Improvement:** B finishes 20ms earlier in HTTP/2.

QUESTION-02

- i. No cache: Total delay = $R = L + A + I$, thus
 $R = 10 + 60 + 180 = 250 \text{ ms}$
- ii. Cache hit ratio $\times$ hit time + Cache miss ratio $\times$ miss time i.e.,
 $R = [0.8 \times (L + P)] + [0.2 \times (L + A + I + P)]$
 $R = 0.8 * 50 + 0.2 * 290 = 98\text{ms}$

**National University of Computer and Emerging Sciences,
Lahore Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	19-February-2026	Section	6A , 6B
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

Q1. A client host in Lahore needs to resolve the IP address for gaia.cs.umass.edu. Assume the following average Round Trip Times (RTT): **[2 + 2 + 1 = 5 Marks][CLO 2]**

- Local DNS Server to Client: 5 ms
 - Local DNS to Root Server: 100 ms
 - Local DNS to .edu TLD Server: 80 ms
 - Local DNS to Authoritative Server: 120 ms
- a) Calculate the total time elapsed from the moment the client sends the DNS query until the IP address is received, assuming an iterative query process and that no records are currently cached.
- b) Suppose the Local DNS server has already cached the IP address of the .edu TLD server from a previous request. Calculate the new total resolution time.
- c) Once the IP is resolved, the client initiates a non-persistent HTTP connection to fetch a base HTML file. If the RTT between the client and the target web server is 150 ms, how much total time passes from the initial DNS query (Part A) until the TCP connection for the HTTP request is fully established?

Q2: A client requests a base HTML file of size 5 KB. This HTML file references **4** very small embedded images (negligible size).

- RTT (Round Trip Time) between client and server is 100 ms.
- Transmission time for the base HTML file is 5 ms.
- Transmission time for the embedded images is negligible (0 ms).
- Assume the TCP connection setup takes 1 RTT.

Calculate the total time elapsed from when the client clicks the link until the base object and all 4 embedded objects are received for the following scenarios:[**4 + 3 + 3 = 10 Marks**][CLO 2]

a) Non-persistent HTTP

b) Persistent HTTP (without pipelining):

c) Persistent HTTP (with pipelining):

SOLUTION QUIZ-02 BSE-6B

QUESTION-01

Part	Formula / Logic	Final Answer
A	$RTT_0 + RTT_{Root} + RTT_{TLD} + RTT_{Auth}$	305 ms
B	$RTT_0 + RTT_{TLD} + RTT_{Auth}$	205 ms
C	Time (Part A) + 1 RTT (TCP Handshake)	455 ms

QUESTION-02

1 RTT = 100 ms.

Base HTML Tx = 5 ms.

Image Tx = 0 ms.

TCP Handshake = 1 RTT.

a. Non-Persistent:

- Base HTML: Handshake (100) + Request/Response (100) + Tx (5) = 205 ms.
- 4 Images: $4 \times (100 + 100 + 0) = 800$ ms. **Total = 1005 ms.**

b. Persistent (No Pipeline):

- Handshake (100) + Request/Response Base (100 + 5) = 205 ms.
- Each Image takes: 1 RTT = 100 ms.
- **Total = 205 + 400 = 605 ms.**

c. Persistent (Pipeline):

- Handshake (100) + Request/Response Base (100 + 5) = 205 ms.
- Client parses, sends 4 requests back-to-back.
- The requests travel together; responses travel together. Total time for all 4 images is 1 RTT.
- **Total = 205 + 100 = 305 ms**

National University of Computer and Emerging Sciences (Lahore Campus)

Quiz 2: Application Layer (Chapter 2)

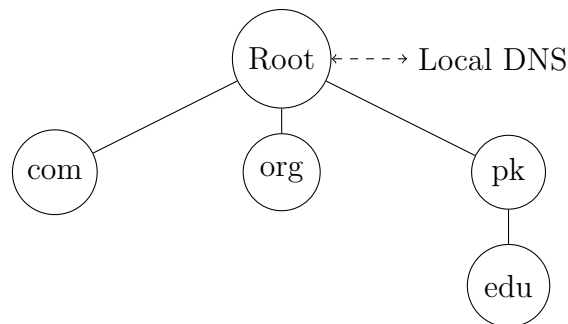
Name: _____ Roll No: _____ Section: BSE-6B1 (Spring 2026)

1. (5 points) DNS Resolution and Latency Calculation

A user requests `www.university.edu.pk`. The local DNS server caches nothing. The delays are:

- Local Host to Local DNS: 2 ms (One way).
- Local DNS to Root Server: 40 ms (RTT).
- Local DNS to .pk TLD Server: 35 ms (RTT).
- Local DNS to .edu.pk Authoritative Server: 20 ms (RTT).
- Local DNS to university.edu.pk Server: 15 ms (RTT).

The DNS resolution is **Iterative**. The RTT between the client and the web server is 50 ms. Assume a new TCP connection is established and requires one RTT for the TCP three-way handshake and one RTT for the HTTP request/response. Calculate the total time from clicking the link to receiving the initial HTTP response, assuming the Local DNS server must contact Root, then TLD, then Authoritative. The local DNS server must also query the final authoritative server for university.edu.pk to obtain the IP address.



2. (10 points) **High-Latency HTTP & Parallel Connections**

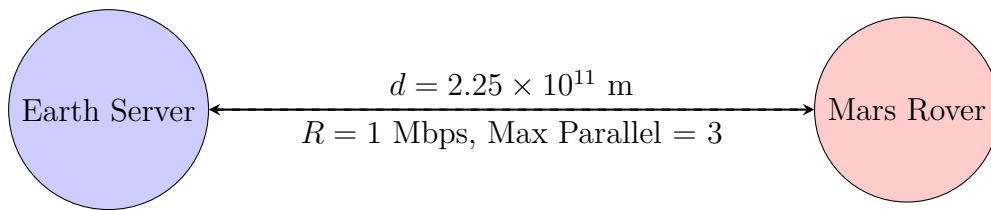
An automated rover on Mars needs to download a firmware update from a server on Earth. The update consists of one base HTML index file (size $L_b = 2$ KB) and 8 referenced binary objects (each size $L_o = 100$ KB).

The link characteristics are:

- Distance Earth-Mars: $d = 2.25 \times 10^{11}$ meters.
- Transmission Rate: $R = 1$ Mbps.
- The rover opens parallel TCP connections but is limited to a maximum of $K = 3$ parallel connections due to memory constraints.

Calculate the total time elapsed from the initial connection request until the last object is fully received. Assume:

- Short control packets (SYN, ACK, HTTP GET) have negligible transmission time compared to propagation delay.
- We are using **Non-Persistent HTTP** over the parallel connections.
- No DNS lookup is required (IP is hardcoded). Speed of light $c = 3 \times 10^8$ m/s.



National University of Computer and Emerging Sciences (Lahore Campus)

Quiz 2: Application Layer (Chapter 2)

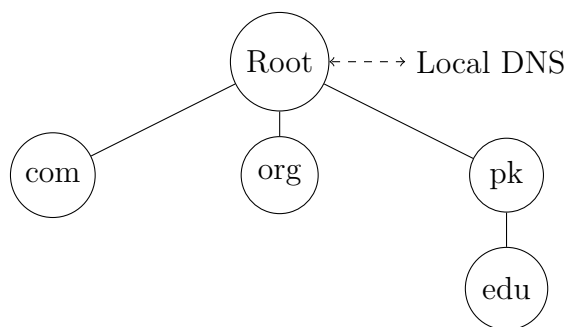
Name: _____ Roll No: _____ Section: BSE-6B1 (Spring 2026)

1. (5 points) DNS Resolution and Latency Calculation

A user requests `www.university.edu.pk`. The local DNS server caches nothing. The delays are:

- Local Host to Local DNS: 2 ms (One way).
- Local DNS to Root Server: 40 ms (RTT).
- Local DNS to .pk TLD Server: 35 ms (RTT).
- Local DNS to .edu.pk Authoritative Server: 20 ms (RTT).
- Local DNS to university.edu.pk Server: 15 ms (RTT).

The DNS resolution is **Iterative**. The RTT between the client and the web server is 50 ms. Assume a new TCP connection is established and requires one RTT for the TCP three-way handshake and one RTT for the HTTP request/response. Calculate the total time from clicking the link to receiving the initial HTTP response, assuming the Local DNS server must contact Root, then TLD, then Authoritative. The local DNS server must also query the final authoritative server for university.edu.pk to obtain the IP address.



Solution:

1. DNS Resolution Time:

$$T_{DNS} = 2 \text{ (Host} \rightarrow \text{DNS)} + 40 \text{ (Root)} + 35 \text{ (TLD)} + 20 \text{ (Auth)} + 15 \text{ (Server)} + 2 \text{ (DNS} \rightarrow \text{Host)}$$

$$T_{DNS} = 114 \text{ ms}$$

2. HTTP Connection & Response:

$$T_{HTTP} = 1 \times RTT_{HTTP} \text{ (TCP Handshake)} + 1 \times RTT_{HTTP} \text{ (HTTP Request)} = 2 \times 50 = 100 \text{ ms}$$

3. Total Time:

$$T_{Total} = 114 \text{ ms} + 100 \text{ ms} = \mathbf{214 \text{ ms}}$$

2. (10 points) **High-Latency HTTP & Parallel Connections**

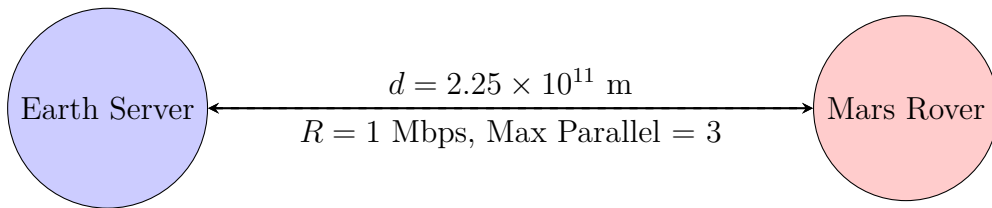
An automated rover on Mars needs to download a firmware update from a server on Earth. The update consists of one base HTML index file (size $L_b = 2$ KB) and 8 referenced binary objects (each size $L_o = 100$ KB).

The link characteristics are:

- Distance Earth-Mars: $d = 2.25 \times 10^{11}$ meters.
- Transmission Rate: $R = 1$ Mbps.
- The rover opens parallel TCP connections but is limited to a maximum of $K = 3$ parallel connections due to memory constraints.

Calculate the total time elapsed from the initial connection request until the last object is fully received. Assume:

- Short control packets (SYN, ACK, HTTP GET) have negligible transmission time compared to propagation delay.
- We are using **Non-Persistent HTTP** over the parallel connections.
- No DNS lookup is required (IP is hardcoded). Speed of light $c = 3 \times 10^8$ m/s.



Solution:

1. Calculations:

$$d_{prop} = \frac{2.25 \times 10^{11}}{3 \times 10^8} = 750 \text{ s} \implies RTT = 2 \times d_{prop} = 1500 \text{ s}$$

$$d_{trans(base)} = \frac{2 \text{ KB} \times 8}{1 \text{ Mbps}} = 0.016 \text{ s}, \quad d_{trans(obj)} = \frac{100 \text{ KB} \times 8}{1 \text{ Mbps}} = 0.8 \text{ s}$$

2. Step 1: Base File Download (Non-Persistent: 2 RTT + Trans)

$$T_{base} = 2 \times RTT + d_{trans(base)} = 3000 + 0.016 = 3000.016 \text{ s}$$

3. Step 2: 8 Objects with K=3 Parallel Connections We download objects in batches. Since $K = 3$, we need $\lceil 8/3 \rceil = 3$ batches. Note: Bandwidth is shared, so transmission time for 3 objects is $3 \times d_{trans(obj)}$.

- **Batch 1 (Obj 1-3):** $2RTT + 3 \times 0.8 = 3000 + 2.4 = 3002.4 \text{ s}$
- **Batch 2 (Obj 4-6):** $2RTT + 3 \times 0.8 = 3000 + 2.4 = 3002.4 \text{ s}$
- **Batch 3 (Obj 7-8):** $2RTT + 2 \times 0.8 = 3000 + 1.6 = 3001.6 \text{ s}$

4. Total Time:

$$T_{total} = 3000.016 + 3002.4 + 3002.4 + 3001.6 = \mathbf{12006.416 \text{ s}}$$